



CS 260 – Foundations of Data Science

Lecture 10 – Information Gain

10/01/2026

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Announcements

HW4 due Friday night

Midterm 1 – October 8th

Office Hours:

Today until 2:15pm



Outline

Information Gain

Continuous Features

Idea of Entropy

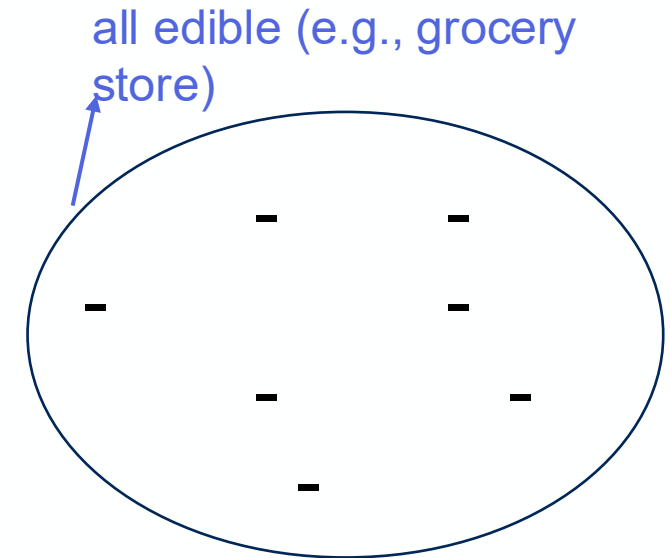
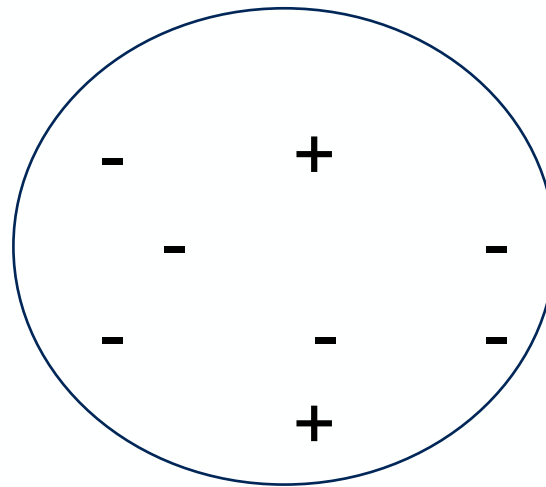
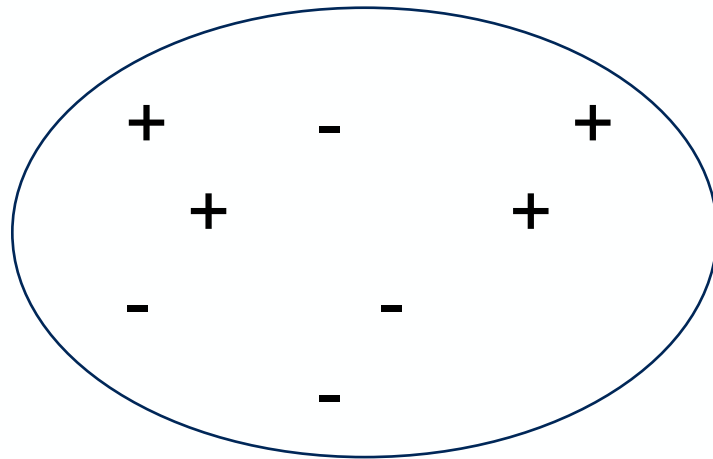
- Average # of bits needed to send one datapoint



Idea of Entropy

- Average # of bits needed to send one datapoint

Poisonous & edible mushrooms



high entropy

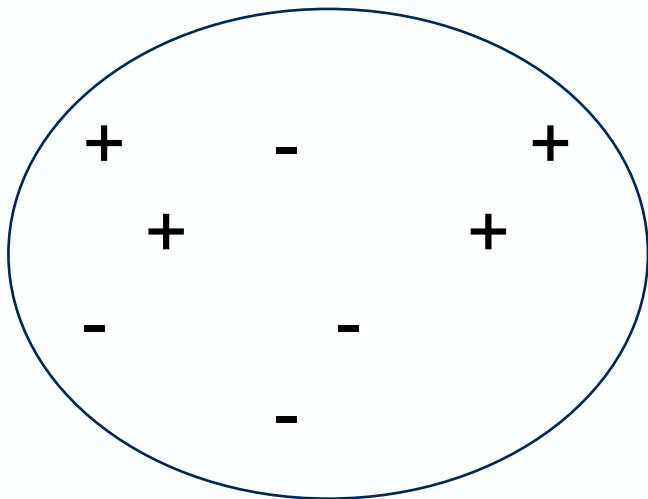
low entropy



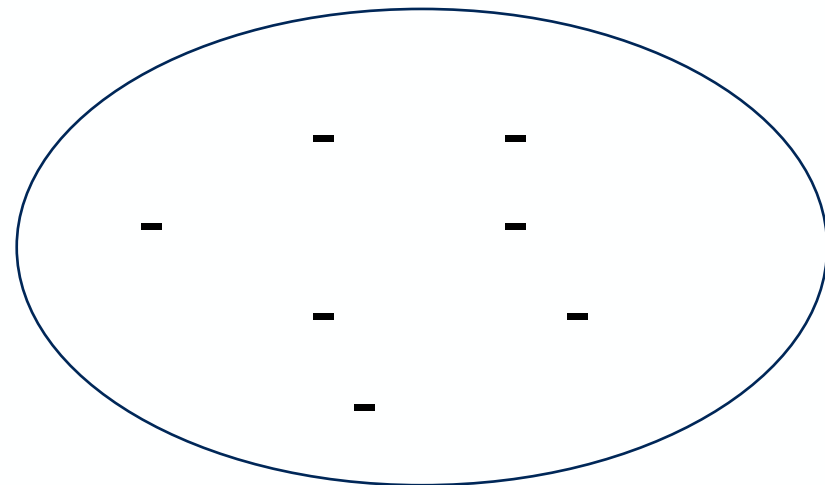
Entropy

$$H(y) = - \sum_{c \in \text{vals}(y)} p(y = c) \underbrace{\log_2(p(y = c))}_{\text{\# of bits}}$$

↑
label



$$H(y) = 1$$



$$H(y) = 0$$



Conditional Entropy

- Quantifies the amount of information needed to describe the outcome of Y given X

$$H(Y|X) = \sum_{v \in \text{vals}(X)} p(X = v) H(Y|X = v)$$

feature

$$H(Y|X = v) = - \sum_{c \in \text{vals}(Y)} p(Y = c|X = v) \log_2 p(Y = c|X = v)$$

single feature value



Information Gain

- Reduction in entropy/uncertainty given some information

$$G(Y, X) = H(Y) - H(Y|X)$$


want high


want low

- Select the feature that maximizes the information gain



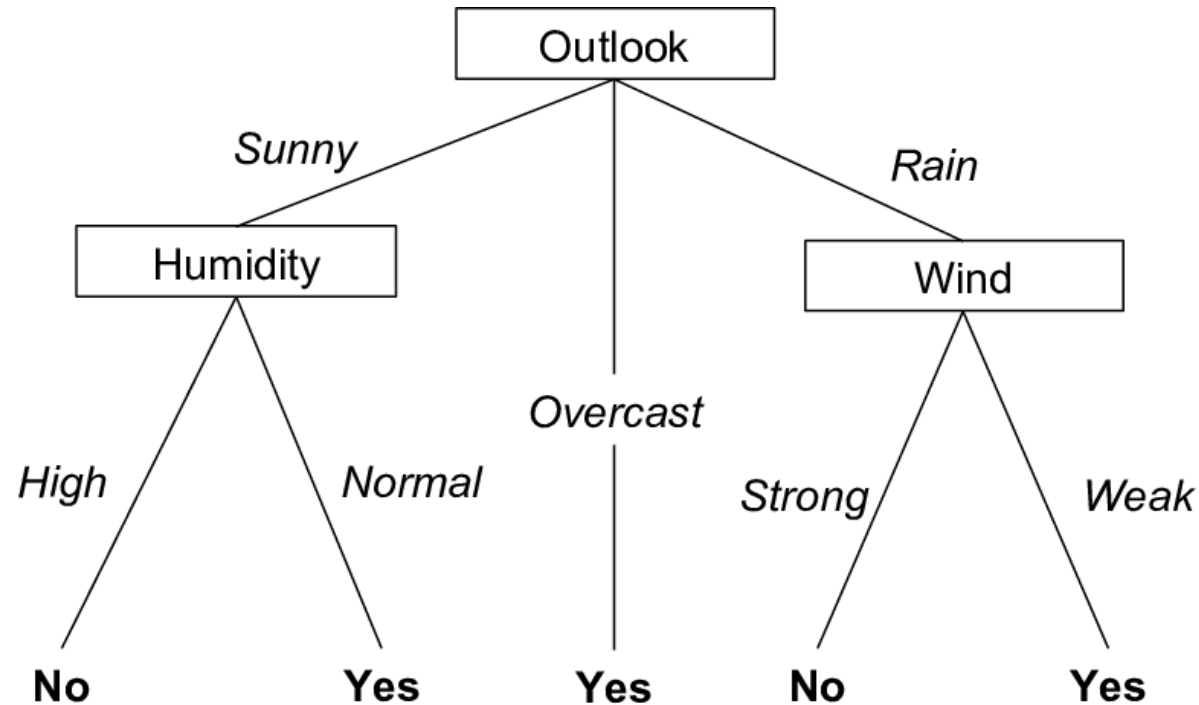
Outline

Information Gain

Continuous Features



Decision Tree example (tennis)



Each internal node: test one feature

Each branch from node: selects one value of the feature

Each leaf node: predict y



Continuous Features

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y



Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature



Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature

2	3	7	7	8	10	12
Y	Y	Y	N	N	Y	Y



Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature

2	3	7	7	8	10	12
Y	Y	Y	N	N	Y	Y

2) Different label with same feature value, collapse to "None"

2	3	7		8	10	12
Y	Y	None	N	Y	Y	



Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature

2	3	7	7	8	10	12
Y	Y	Y	N	N	Y	Y

2) Different label with same feature value, collapse to “None”



Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature

2	3	7	7	8	10	12
Y	Y	Y	N	N	Y	Y

2) Different label with same feature value, collapse to “None”

2	3	7		8	10	12
Y	Y	None		N	Y	Y

3) Whenever label changes, make a feature (use avg)

2	3	7		8	10	12
Y	Y	None		N	Y	Y

Continuous Features

(do this for the TRAIN only!)

X	Y
10	Y
7	Y
8	N
3	Y
7	N
12	Y
2	Y

1) Sort examples based on given feature

2	3	7	7	8	10	12
Y	Y	Y	N	N	Y	Y

2) Different label with same feature value, collapse to "None"

2	3	7		8	10	12
Y	Y	None		N	Y	Y

3) Whenever label changes, make a feature (use avg)

2	3	7		8	10	12
Y	Y	None		N	Y	Y

$X \leq 5$ $X \leq 7.5$ $X \leq 9$



Continuous Features (Handout 13)

(do this for the TRAIN only!)

temp	Y
80	Y
48	Y
60	N
48	N
40	N
48	Y
90	Y

1) Sort examples based on feature “temp”

2) Different label with same feature value, collapse to “None”

3) Whenever label changes, make a feature (use avg)



Continuous Features (Handout 13)

(do this for the TRAIN only!)

temp ≤ 44	temp	Y
F	80	Y
F	48	Y
F	60	N
T	48	N
F	40	N
F	48	Y
F	90	Y

↑
3 new
columns

1) Sort examples based on feature “temp”

40	48	48	48	60	80	90
N	Y	N	Y	N	Y	Y

2) Different label with same feature value, collapse to “None”

40	48	60	80	90
N	None	N	Y	Y

3) Whenever label changes, make a feature (use avg)

40	48	60	80	90
N	None	N	Y	Y

X ≤ 44

X ≤ 54

X ≤ 70

